

# NATHAN HAY

Clarkson, WA 6030 | 0444 588 636 | nhay0970@gmail.com | [GitHub Portfolio](#)

## PROFESSIONAL SUMMARY

Motivated IT/Cybersecurity university student and AAFC 721 Squadron cadet with a strong foundation in defensive cybersecurity, leadership, and operational discipline. Actively building a comprehensive, evidence-based portfolio of practical cybersecurity projects (homelab architecture, penetration testing, SIEM, digital forensics, and secure web application development) aligned with ASD Essential Eight and ISM frameworks. Passionate about drone operations, UAS reconnaissance, and information operations. Preparing for enlistment as a Navy Information Warfare Officer while pursuing long-term goals in defence and government information/cyber roles. Demonstrated ability to lead teams, manage complex technical projects, and communicate effectively under pressure.

## CORE COMPETENCIES

|                           |                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cybersecurity & Technical | Network Segmentation & Hardening (pfSense) • Penetration Testing & Red Teaming • SIEM Deployment & Detection Engineering (Wazuh) • Digital Forensics (Autopsy + Volatility) • Secure Coding & Application Security (Flask/OWASP) • Post-Quantum Cryptography Concepts • Drone Operations & UAS Reconnaissance |
| Leadership & Operations   | Team Leadership & Training (30+ cadets) • Operational Planning & Execution • Stakeholder Communication • Risk Assessment & Safety Compliance • Mentoring & Performance Improvement                                                                                                                            |

## CYBERSECURITY PROJECTS PORTFOLIO

Full Portfolio & Detailed Reports: [github.com/Nathanthegunner0/raaf-cyber-portfolio](https://github.com/Nathanthegunner0/raaf-cyber-portfolio)

- Cybersecurity Home Lab Design** — Designed and implemented a segmented multi-VM defensive network (pfSense firewall, Active Directory, Metasploitable targets, Kali attacker) with full Essential Eight mapping and professional documentation.
- Penetration Testing & Essential Eight Remediation** — Conducted full authorised pentest on lab environment, exploited vulnerabilities (vsftpd, SMB, SQLi, file upload), produced professional report, and implemented verified remediations mapped to all 8 Essential Eight controls.
- SIEM Implementation & Threat Hunting (Wazuh)** — Deployed centralised Wazuh SIEM across all lab assets, developed custom detection rules, validated real-time alerts on live attacks from Project #2, with MITRE ATT&CK; mapping and dashboard creation.
- Digital Forensics & Incident Response** — Performed complete forensics simulation on compromised VM: acquired E01 disk + memory images, analysed with Autopsy & Volatility, reconstructed full attack timeline, and developed practical response playbooks.
- Secure Web Application Development & Hardening** — Built production-grade Flask web app from scratch with secure auth, input sanitisation, security headers, SIEM logging integration, and full OWASP Top 10 mitigations. Achieved clean ZAP scan after remediation.

## RELEVANT EXPERIENCE

### Australian Air Force Cadets — 721 Squadron (WA)

Second in Command & Field Craft Leader | Kingsley, WA | Feb 2021 – Nov 2024

- Led training, welfare, and operational readiness for 30+ cadets during high-intensity military exercises, achieving 100% safety compliance and significantly improved team performance under pressure.
- Coordinated daily communications between 50+ cadets, parents, and staff, reducing miscommunications by 30% and improving overall operational efficiency.
- Conducted weekly morale and performance check-ins with 15+ cadets, resulting in a 20% increase in team engagement scores over three months.
- Planned and executed synchronized marching and field craft training for teams of up to 50 cadets, contributing to a 15% improvement in parade and operational performance scores.

### Drone Operations & Reconnaissance Experience

Drone Pilot & UAS Operator | Personal & Training Projects | 2023 – Present

- Operated multi-rotor drones (DJI and similar platforms) for aerial surveillance, mapping, and reconnaissance exercises, maintaining full compliance with CASA regulations and safety protocols.
- Conducted pre-flight risk assessments, flight planning, and post-flight data analysis, integrating drone imagery with digital tools to support situational awareness and project-based applications.

- Developed practical experience in drone data collection and basic analysis techniques relevant to security, infrastructure monitoring, and emergency response scenarios.

## EDUCATION & CERTIFICATIONS

---

|                                                                               |                      |
|-------------------------------------------------------------------------------|----------------------|
| CISCA High School Certificate                                                 | Graduation: Nov 2024 |
| RSA – Responsible Service of Alcohol (AHA WA)                                 | Feb 2025             |
| Manual Driver's Licence (Western Australian Transport Services)               | Nov 2023             |
| Current: University Studies in IT/Cybersecurity (strong academic performance) | Ongoing              |

*Full project reports and evidence available in the public GitHub portfolio linked above. Prepared with a focus on practical, demonstrable skills aligned with ADF cyber requirements and the Essential Eight framework.*